

Piyush Maiti

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Education

M.S., Biomedical Engineering — University of Southern California, Los Angeles (Dec 2018)

B.E., Electrical & Electronics Engineering — Bhilai Institute of Technology, India (Aug 2014)

Research Experience

Neuroimaging Data Analyst, Memory and Aging Center, UCSF (Apr 2023 – Present)

- Key member of the PET Core for the Longitudinal Early-onset Alzheimer's Disease Study (LEADS); responsible for processing and analyzing multimodal PET/MRI data across sites, managing large-scale imaging datasets, developing data extraction tools, and performing statistical analyses based on study requirements.
- Developed automated quality control (QC) tools for MRI (SPM/FreeSurfer) and PET pipelines, deployed across projects.
- Developed analytic workflows to identify participants on anti-amyloid therapies and quantify longitudinal tracer effects; contributed to analyses of target engagement and cognitive outcomes in early-onset Alzheimer's disease (LEADS cohort).
- Contributed to analyses of discordance between amyloid PET visual reads and quantitative measures; performed statistical and image-based analyses in LEADS and Alzheimer's Disease Neuroimaging Initiative (ADNI) for [^{18}F]-Florbetaben, [^{18}F]-Florbetapir, and [^{18}F]-Flutemetamol amyloid tracers.
- Processed and harmonized MRI-based amyloid PET data using FreeSurfer- and SPM-based pipelines to extract Centiloid quantifications, supporting cross-cohort analyses of amyloid pathology differences between early- and late-onset Alzheimer's disease (EOAD vs. LOAD).
- Evaluated WMH segmentation reliability and FreeSurfer-based PET quantification variability to improve signal reproducibility.
- Mentored research assistants and supported neurologists with participant-level PET/MRI interpretation reports.

Staff Scientist, Mallinckrodt Institute of Radiology, Washington University School of Medicine (Jul 2021 – Mar 2023)

- Evaluated mouse brain ROI segmentation tools using the MOBY digital phantom; validated region-wise tracer uptake and developed a 3D PET quantification framework to minimize partial volume effects and enhance quantitative accuracy.
- Worked on optimization and harmonization of PET scanner reconstructions for partial volume effects using Phantech's linear-filling phantom with [^{18}F]-FDG and [^{64}Cu] isotopes, comparing OSEM3D/MAP and TeraTomo 3D reconstructions across varying voxel sizes, iterations, and energy windows.
- Developed a PET-CT preprocessing pipeline for a longitudinal mouse brain study to identify imaging biomarkers of obesity and Alzheimer's disease, involving brain registration, SUV extraction, and kinetic modeling using [^{18}F]-FSPG radiotracer.
- Developed a cross-modality radiomics analysis tool to correlate features between histopathology (Nissl stain), autoradiography, and PET/MRI data in mouse brain and tumor imaging; implemented advanced image registration and computational methods to integrate radiomic features across modalities.

- Simulated compartmental-model-based time-activity curves to study sampling bias and optimize PET acquisition protocols.

Project Assistant, Imaging Genetics Center, USC Keck School of Medicine (Jul 2019 – Jun 2021)

- Assisted in developing deep learning methods for scanner invariance and data harmonization across multi-site MRI datasets; characterized feature disparities among images from different scanners and acquisition protocols, and implemented GAN-based style transfer to enhance cross-site consistency for clinical analysis.
- Designed preprocessing protocols for ADNI-3 diffusion MRI, developing and testing image processing tools and conducting statistical analyses to evaluate denoising methods across scanner vendors and sites. Created interactive HTML-based platforms for quality control, data visualization, and extraction.
- Developed tools to quantify Gibbs artifacts in MRI using the Physically Implausible Signal framework; generated simulated Gibbs distortions on QSM data for U-Net–based deep learning artifact removal.
- Built auto-mask correction tools for EPI distortion and temporal/optic nerve exclusion using edge-based filtering.
- Developed automated masking and QC tools for structural and diffusion-weighted MRI, including a T1/DTI-based tissue classifier for EPI correction, Gaussian/Sobel-based masking of optic nerves and temporal lobes integrated into ADNI pipelines, and tested ASHS (Automatic Segmentation of Hippocampal Subfields) for hippocampal segmentation and QC of high-resolution 7T imaging.

Research Assistant, Computational Imaging of Brain Organization Lab, Children’s Hospital Los Angeles (Oct 2018 – Jun 2019)

- Built automated diffusion MRI pipelines for neonatal brain imaging to assess white-matter injury; reduced manual QC effort by 30%.

Technical Skills

Programming: Python, MATLAB, C++, HTML | Scripting: Bash, Batch

Libraries/Tools: Nilearn, NumPy, OpenCV, Numba, Dask, MySQL

Imaging Tools: SPM, FSL, FreeSurfer, DIPY, MRtrix, PMOD, ITK-Snap, BrainSuite, VivoQuant

Publications & Abstracts

Papers and Publications

- Lagarde J., **Maiti P.**, Schonhaut D. R., Blazhenets G., Zhang J., Eloyan A., Thangarajah M., et al. *Amyloid PET in Sporadic Early-Versus Late-Onset Alzheimer’s Disease: Comparison of the LEADS and ADNI Cohorts.* *Annals of Neurology*, 2025.
- Zhang J., Soleimani-Meigooni D. N., Koeppe R., Ackley S., Schonhaut D. R., Lin Z. E., **Maiti P.**, et al. *Exploring Centiloid Robustness: Impact of Sample Size and Image Resolution on Centiloid Conversion Accuracy.* *Journal of Nuclear Medicine*, 2025.
- Lin M., Chiotis K., **Maiti P.**, Zhang J., Blazhenets G., et al. *Tau Topography Subtypes Account for Clinical Heterogeneity and Longitudinal Trajectories in Early-Onset Alzheimer’s Disease.* *Brain Communications*, under review.
- Guo Z., Luo J., Mashl R. J., Hoog J., **Maiti P.**, Fettig N., Davies S. R., et al. *Evaluation of Copanlisib in Combination with Eribulin in Triple-Negative Breast Cancer Patient-Derived Xenograft Models.* *Cancer Research Communications*, 4(6), 2024: 1430–1440.

- Liu M., **Maiti P.**, Thomopoulos S., Zhu A., Chai Y., Kim H., Jahanshad N. *Style Transfer Using Generative Adversarial Networks for Multi-Site MRI Harmonization*. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 313–322. Cham: Springer International Publishing, 2021.
- Thomopoulos S., Nir T. M., Villalon-Reina J. E., Zavaliangos-Petropulu A., **Maiti P.**, Zheng H., Nourollahimoghadam E., et al. *Diffusion MRI Metrics and Their Relation to Dementia Severity: Effects of Harmonization Approaches*. *International Symposium on Medical Information Processing and Analysis (ISPA)*, Brazil, 2021.
- Gadewar S., Zhu A. H., Thomopoulos S. I., Li Z., Ba Gari I., **Maiti P.**, Thompson P. M., Jahanshad N. *Region-Specific Automatic Quality Assurance for MRI-Derived Cortical Segmentations*. *IEEE International Symposium on Biomedical Imaging (ISBI)*, France, 2021.

Conference Abstracts

- Heston M. B., **Maiti P.**, Blazhenets G., Soleimani-Meigooni D. N., Lin Z., et al. *Discordance Between Amyloid PET Visual Read and Quantitation: False Positives or Early Pathology Detection? Human Amyloid Imaging (HAI)*, 2026 (under review).
- Rocha S., Pasternak S., Lesman-Segev O., Schwartz D., Blazhenets G., **Maiti P.**, et al. *Amyloid-PET and White Matter Hyperintensities but Not Military Service Duration nor Combat Exposure Are Associated with Neurodegeneration and Cognitive Impairment in Older Veterans*. *Human Amyloid Imaging (HAI)*, 2026 (under review).
- Ziontz J., **Maiti P.**, Zhang J., Chiotis K., Blazhenets G., Rocha S., Shankar R., et al. *Clinical and Cognitive Manifestations of Hemispheric Asymmetry in Neuroimaging Biomarkers in Early-Onset Alzheimer's Disease*. *Alzheimer's Association International Conference (AAIC)*, PA, 2025.
- Chiotis K., Blazhenets G., Schonhaut D. R., Lagarde J., **Maiti P.**, Zhang J., Shankar R., et al. *Tau PET Load in Early- and Late-Onset Alzheimer's Disease: A Cross-Sectional and Longitudinal Comparison of the LEADS and ADNI Cohorts*. *Alzheimer's Association International Conference (AAIC)*, 2025.
- Njamnshi W. Y., Chiotis K., Tsoy E., Possin K. L., Hammers D. B., Eloyan A., Kirby K., **Maiti P.**, et al. *Tablet-Based Cognitive Assessment in a Multisite Study of Early-Onset Alzheimer's Disease: Association with Pen-and-Paper Tests and PET Measures of Amyloid and Tau*. *Alzheimer's Association International Conference (AAIC)*, 2025.
- Schonhaut D. R., Insel P. S., Soleimani-Meigooni D. N., Zeltzer E., Windon C., **Maiti P.**, Yballa C., et al. *A Clock Model of Regional Tau-PET Accumulation and Cognitive Decline in Sporadic Early-Onset Alzheimer's Disease*. *Alzheimer's & Dementia*, 20 (2024): e093935.
- Maiti P., Li S., Dutta K., Whitehead T., Shoghi K. *Assessing the Performance of Software Platforms to Segment Mouse Brains Using a Digital Phantom*. *SNMMI 2023*, Chicago, IL.
- Shyam C., Whitehead T., Zhou D., Qiu L., Maiti P., Tu Z., Kravitz A., Hershey T., Shoghi K. *Imaging Neuroinflammation in an Animal Model of Diet-Induced Obesity*. *SNMMI 2023*, Chicago, IL.
- Liu M., Maiti P., Thomopoulos S., Kadakova N., Kim H., Jahanshad N. *Multi-Site MRI Harmonization Using Unified Generative Adversarial Networks*. *OHBM 2021*, Seoul, Korea.
- Ba Gari I., Gadewar S. P., Wei X., Maiti P., Boyd J., Jahanshad N. *A 3D U-Net for Gibbs Artifact Removal from Quantitative Susceptibility Maps*. *ISMRM 2021*, Chicago, IL.
- Nir T. M., Salminen L., Villalon-Reina J. E., Tubi M. A., Thomopoulos S. I., **Maiti P.**, Braskie M. N., Thompson P. M., Jahanshad N. *Hippocampal Subfield Microstructure Abnormalities Mediate Associations Between Tau Burden and Memory Performance: Neuroimaging/Multi-Modal Comparisons*. *Alzheimer's & Dementia*, 16 (2020): e039622.

- Nir T. M., Villalon-Reina J. E., Thomopoulos S. I., Maiti P., Reid R., Bernstein M., Jack C. R., Thompson P. M., Jahanshad N. *Improved Neurite Density Estimation in Alzheimer's Disease With Multi-Tissue Multicompartment Diffusion MRI Modeling*. IEEE EMBC 2020, Montreal, Canada.
- Nir T. M., Villalon-Reina J. E., Zhu A., Salminen L. E., Thomopoulos S. I., Tubi M. A., Maiti P., Thompson P. M., Jahanshad N. *Hippocampal Microstructural Abnormalities in Cognitively Impaired and Amyloid-Positive Individuals*. OHBM 2020, Montreal, Canada.

Journal Review

Reviewer — IEEE International Symposium on Biomedical Imaging (ISBI)